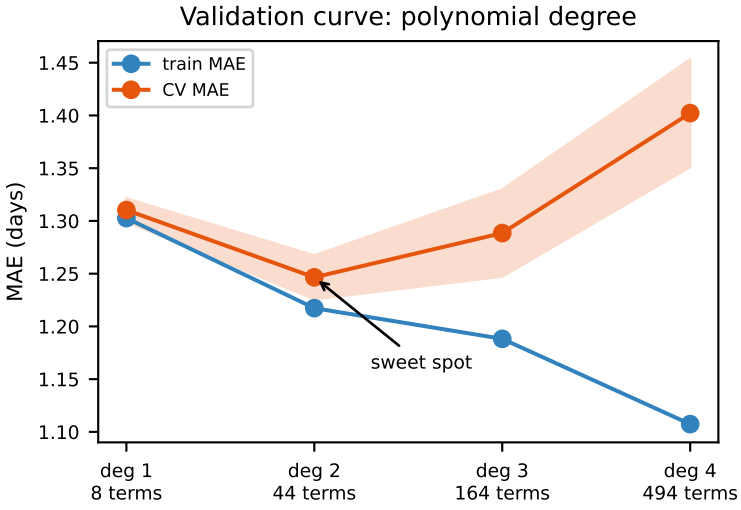
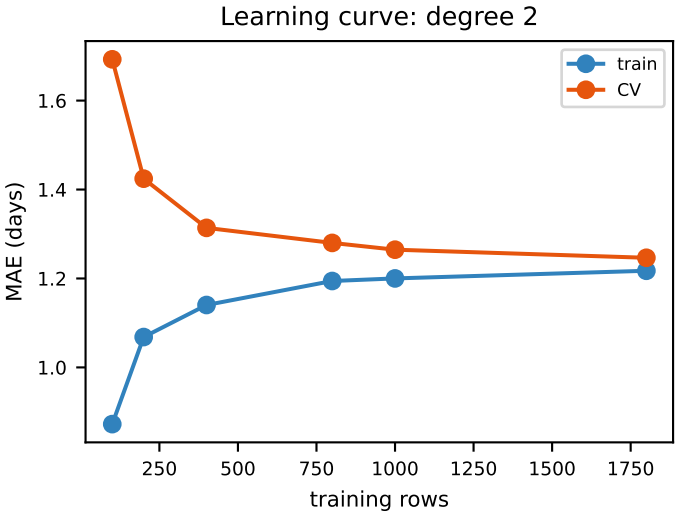
Day 26 Cheat Sheet: Learning and Validation Curves

Week 4: diagnose under- and overfitting before you tune anything

TERM	WHAT IT MEANS	CODE
Training error	error on rows the model was fit on (optimistic)	-tr.mean(1)
CV error	error on held-out folds inside TRAIN (honest)	-va.mean(1)
Gap	CV error minus train error; big gap = variance	cv_mae - train_mae
High bias (underfit)	train AND CV error high, gap small	degree 1 here
High variance (overfit)	train low, CV high, gap large	degree 4 here
Validation curve	error vs ONE knob (degree, alpha)	validation_curve(...)
Learning curve	error vs training size, model fixed	learning_curve(...)
Plateau	curves flat and close: more data will not help	add features instead
Converging gap	gap still closing: more rows could help	collect more data / regularize





THIS DATA: 3,000 simulated discharges (seed 26), 8 features, continuous los_days, 75/25 split (train 2,250, test 750)
Validation curve (5-fold CV MAE): deg1 1.310 | deg2 1.247 | deg3 1.288 | deg4 1.402
Gap by degree: 0.008 | 0.029 | 0.100 | 0.295. Terms: 8, 44, 164, 494.
Degree 4 with alpha 0.001 scores CV 1.402; with alpha 316 it scores 1.262.
Champion by CV: degree 2. Test once: MAE 1.254 days, RMSE 1.570, R2 0.774.

READING THE CURVES
- Both curves high and close together: underfit. Add features, interactions or a more flexible model. More rows will not help.
- Train low, CV high, big gap: overfit. Simplify, regularize (Day 23) or add data.
- The best setting is where CV error bottoms out, not where train error is lowest.
- Train error always falls as the model grows. Only the held-out curve can tell you the truth.
- Never tune on the test set. Curves use CV inside the training set; test is scored once at the end.

```
tr, va = validation_curve(make(1), X_train, y_train,
    param_name="polynomialfeatures__degree", param_range=[1, 2, 3, 4],
    cv=cv, scoring="neg_mean_absolute_error")
```

Infographic: Too simple, just right, too flexible

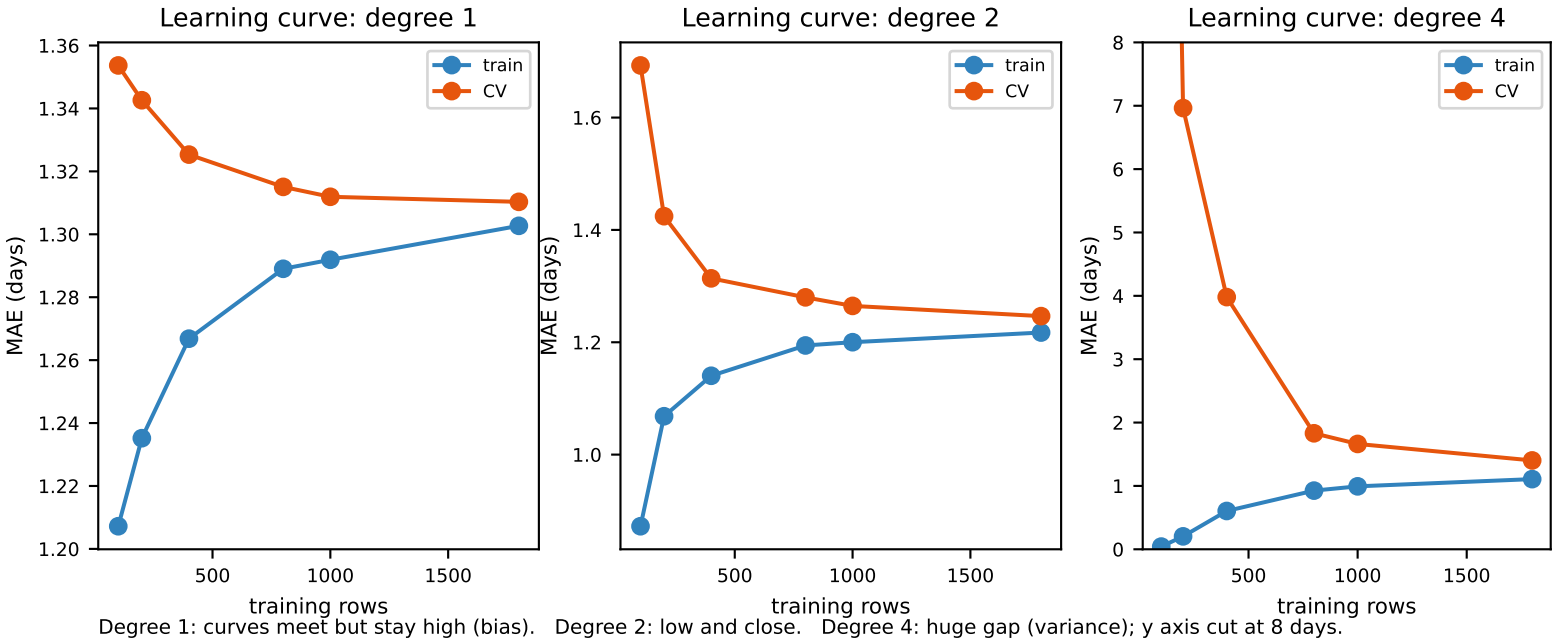
Same data, three degrees of flexibility

DEG 1 GAP
0.008 d

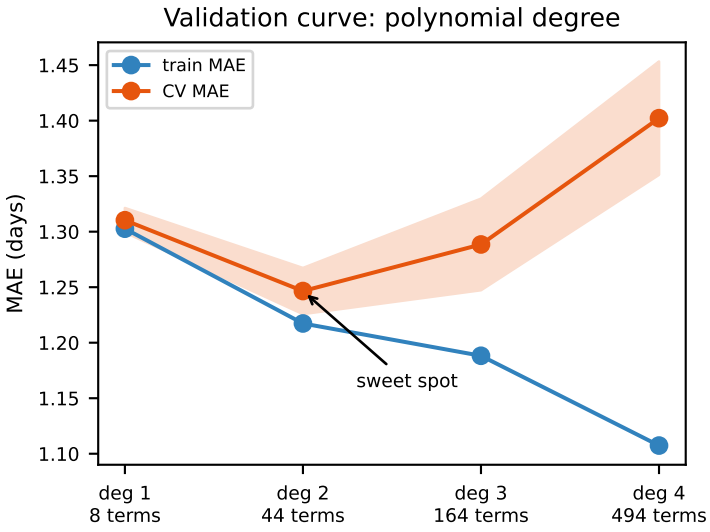
DEG 2 CV MAE
1.247 d

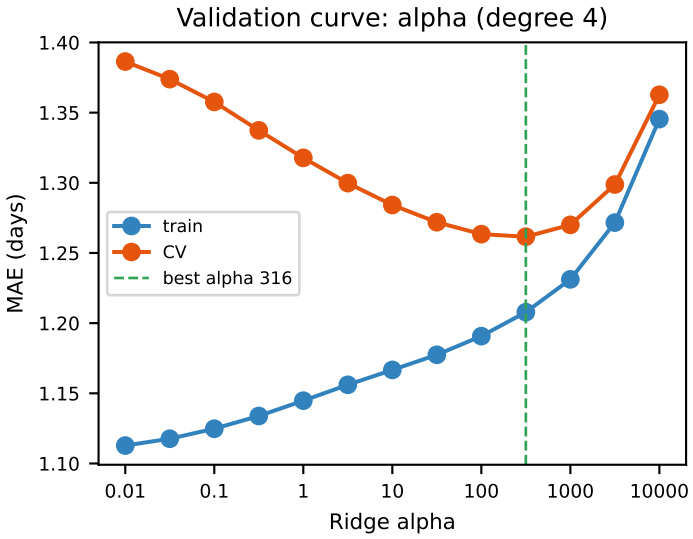
DEG 4 GAP
0.295 d

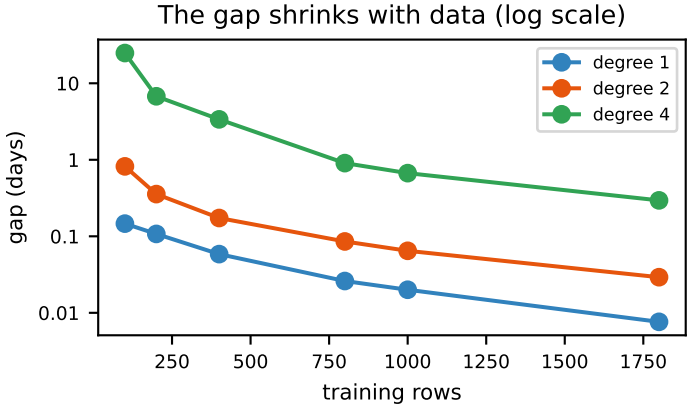
TEST MAE
1.25 d

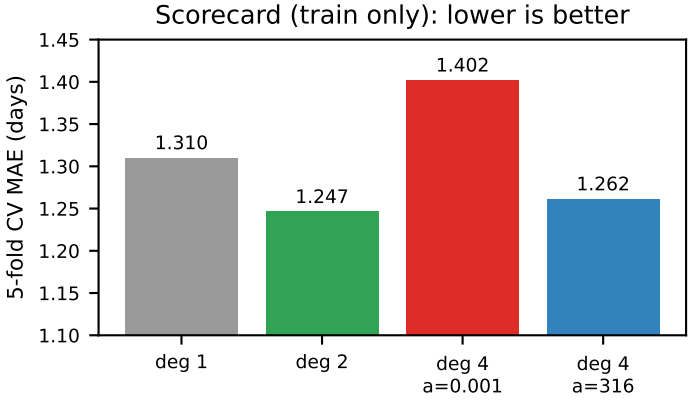


Degree 1: curves meet but stay high (bias). Degree 2: low and close. Degree 4: huge gap (variance); y axis cut at 8 days.









Hands-On Worksheet: ds-genai-90-days / day26

60-90 minutes | notebook: day26/curves_los.ipynb + day26/README.md

[ ] mkdir day26, conda activate ds90, open day26/curves_los.ipynb

[ ] Simulate ~3,000 discharges with interactions and age curve (snippet 1, seed 26)

[ ] Split 75/25; build a 5-fold KFold on TRAIN only (snippet 2)

[ ] Write a make(degree, alpha) pipeline: poly -> scale -> Ridge

[ ] Run validation_curve over degree 1 to 4; print train, CV and gap (snippet 2)

[ ] Count terms per degree: 8, 44, 164, 494

[ ] Write the diagnosis rule BEFORE test: gap > 0.15 days = variance

[ ] Run learning_curve for degree 1, 2 and 4 (snippet 3)

[ ] Label each: bias, good fit, or variance. Does the gap still close?

[ ] Validation curve over alpha for degree 4 (snippet 4); note best alpha

[ ] Build the CV scorecard of four candidates; pick champion by CV only

[ ] Score test ONCE: MAE, RMSE, R2 (snippet 5)

[ ] README: three curve plots, diagnosis, champion, limits (simulated data)

[ ] git add day26 && git commit && git push

[ ] Stretch: Snowflake SQL to log curve results per run (snippet 7)

```
pipe = make_pipeline(PolynomialFeatures(degree, include_bias=False),
                     StandardScaler(), Ridge(alpha=alpha))
n, tr, va = learning_curve(pipe, X_train, y_train, train_sizes=sizes,
    cv=cv, scoring="neg_mean_absolute_error", shuffle=True, random_state=26)
gap = (-va.mean(1)) - (-tr.mean(1))
```

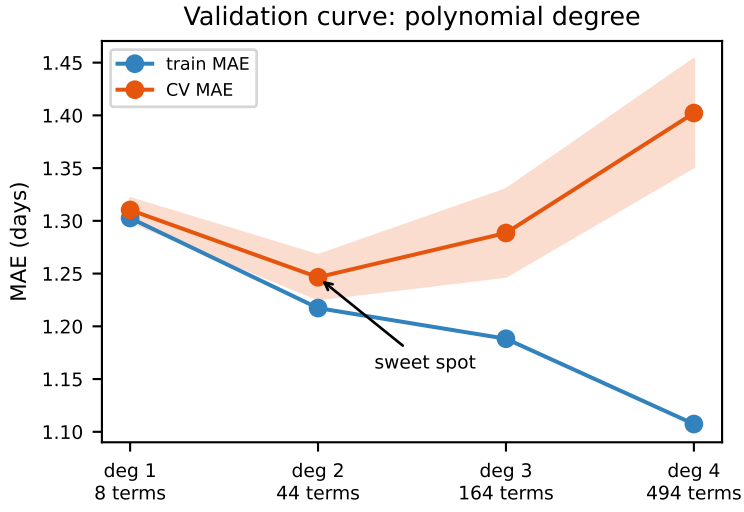
CV MAE deg1: _____ deg2: _____ deg3: _____ deg4: _____

Gap deg1: _____ deg2: _____ deg3: _____ deg4: _____

Best alpha (deg 4): _____ Champion: _____

Test MAE: _____ d RMSE: _____ R2: _____

Notes: _____



Day 26 Self-Check Quiz

1. Train MAE 0.4 days, CV MAE 2.1 days. Bias or variance? What would you try?

_____

2. Train MAE 1.30 and CV MAE 1.31, both far above what you need. Diagnosis?

_____

3. Learning curve gap is still closing at the largest size. What does that suggest?

_____

4. What is the difference between a validation curve and a learning curve?

_____

5. Why must the curves be computed on the training set only?

_____

6. Why does train error keep falling as polynomial degree grows?

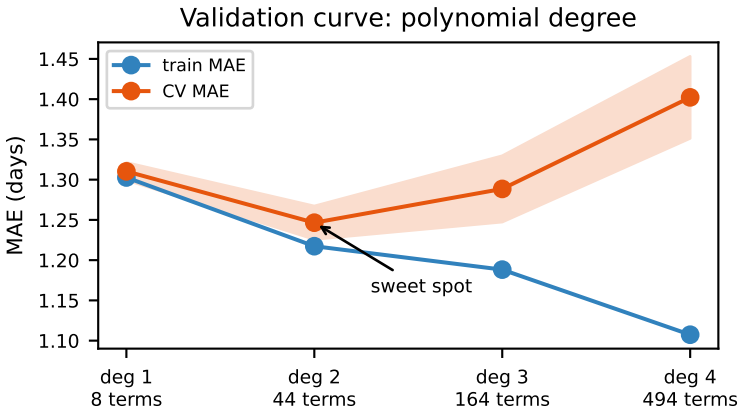
_____

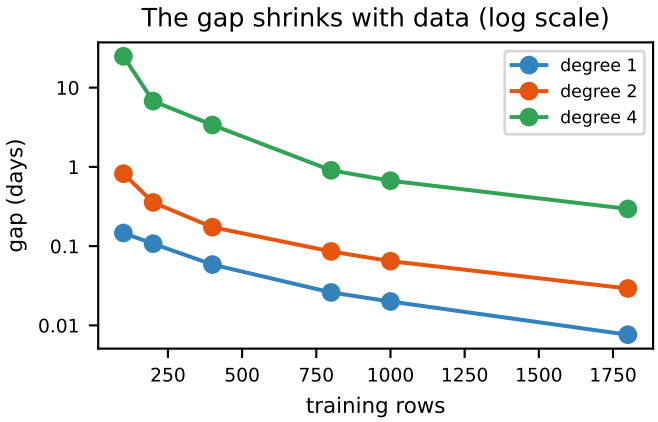
7. Degree 4 gave 494 terms on 2,250 rows. How did Ridge alpha help?

_____

8. Today's champion was degree 2. Why not degree 1, which had the smallest gap?

_____





Answers:
1 variance (overfit): simplify, regularize or add data
2 bias (underfit): add features, interactions or flexibility; more rows will not help
3 more rows could still help (the model is variance limited)
4 validation curve varies a model knob at fixed data; learning curve varies training size at a fixed model
5 test must stay untouched until the single final score, or it stops being honest
6 more terms can always fit the training rows better, including noise
7 the penalty shrinks coefficients: CV MAE went from 1.402 to 1.262
8 gap is not the goal: deg 1 CV MAE 1.310 is worse than deg 2 at 1.247; pick the lowest CV error